

Francisco Alfonso Mayorga Cetina

Mathematician | Software Developer | Development Leader | Data Engineer

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Summary

Mathematics professional with over five years of experience in software development using .NET technologies (C#, F#, Visual Basic), desktop applications (WPF), and data tools (Python, SQL).

Proven track record leading development teams and delivering high-quality solutions. Experience in backend development, desktop user interfaces, cloud platforms, and API integration.

Currently seeking roles that allow greater involvement with modern .NET technologies, frontend frameworks, and cloud-native architectures.

Technology Stack

- **Backend:** C# (.NET Framework, migrating to .NET 8+), Entity Framework, REST APIs, WCF Services
 - **Frontend:** WPF (MVVM), DevExpress XAF, Basic React
 - **Cloud:** Azure (Entra ID, App Registrations, Azure SQL), AWS (EC2, S3, Load Balancer), GCP (Cloud Functions, BigQuery)
 - **DevOps & Infrastructure:** Azure DevOps, Terraform, Git, SonarQube, Redmine, Jira, Confluence
 - **Databases:** SQL Server, Oracle (PL/SQL), MySQL
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Experience

Universidad Sergio Arboleda

Skills & Tools: LaTeX, SAP, Office, Python (NumPy, Pandas, Matplotlib, Pygame, etc.), Jupyter, AI Tools (GitHub Copilot, Chat GPT, Gemini, etc.)

Adjunct Professor (Oct 2025 – Dec 2025) – Scientific Computing

- Teach theoretical and practical programming classes applied to mathematics.
- Evaluate and accurately record grades in SAP.

Finanzas y Actuaría (Finac S.A.S)

Skills & Tools: Visual Studio, Azure DevOps (TFS-Git), .NET Framework (C#, F#, Visual Basic), SQL Databases, Redmine, SonarQube, DevExpress, ADO.NET, WPF, Problem Solving, Leadership

Development Coordinator – Investments Team (Jan 2025 – Present)

- Lead the development of products related to investment instruments.
- Coordinate new cloud-based deployments and integration with enterprise identity systems.
- Provide customer-facing support and incident resolution.

Development Coordinator – Support and Maintenance (Sep 2023 – Jul 2024)

- Supervise the maintenance development team.
- Manage resource allocation and task tracking through Azure DevOps.
- Lead software improvements based on WPF and .NET.

Development Professional II (Jan 2023 – Aug 2023)

- Guide junior developers and ensure timely project delivery.
- Develop WPF applications aligned with business needs.

Development Analyst (Nov 2020 – Dec 2022)

- Develop and maintain WinForms and ASP.NET applications.
- Design and develop REST and SOAP APIs.
- Work with SQL Server and Oracle databases.

CALA Analytics (By Infórmese)

Skills & Tools: Google Cloud Platform, Python, React, GitLab, Terraform, Confluence

Analytics Consultant (Jul 2024 – Dec 2024)

- Build scalable solutions on GCP using Python and BigQuery.
- Design CI/CD pipelines and report automation.
- Use Terraform for infrastructure provisioning.

Sii Group Colombia (Thales Project)

Skills & Tools: Visual Studio, Mercurial (GIT), .NET Framework, WPF, API Integration, Confluence

Senior .NET Developer (Aug 2023 – Sep 2023)

- Maintain WPF applications for a global security platform.
- Integrate and document a set of APIs.

Universidad Sergio Arboleda

Skills & Tools: Office, Leadership, Assertive Communication, Mathematics, Statistics, Moodle

Teaching Assistants Coordinator (Jun 2018 – Jan 2020)

- Team member selection processes.
- Evaluation of results at the end of academic period.
- Logistics in integration activities.

Continuing Education Instructor (Mathematical Talents Program) (Mar 2019 – May 2019)

- Education for children and young people with aptitude for mathematics.
- Strengthening skills for the dissemination of scientific knowledge.

Academic Teaching Assistant (Mathematics Dept.) (Jan 2016 – Dec 2019)

- Support students in developing class activities.
- Teaching assistant for grading and other operational tasks.

Fundación Haiko

Volunteer (Feb 2019 – Nov 2020)

- Support teachers and students during academic activities.
- Assist in logistical tasks for the development of various classes.

Education

Mathematics – Universidad Sergio Arboleda (2014–2020)

Honors Thesis: Characterizing Dual Graphs Associated with Triangulations of the Caracol Flow Polytope.

Courses and Certificates

- Exploring your DATAVERSE with IBM SPSS Modeler - [Cala Academy](#), Bogotá, Colombia (Jul 2024)
- B2 Upper Intermediate - [EF SET](#) English Certificate (Aug 2023)
- HTML, JavaScript and Services – [ITTalent](#), Bogotá, Colombia (Aug 2022)
- UI/UX Certification Course – [Programming Hub](#) (Nov 2021)
- Python 3 Certification – [Programming Hub](#) (Aug 2020)
- Advanced Python Course – [Programming Hub](#) (Aug 2020)
- Ethical Hacking with Python – [Udemy](#) (Feb 2020)

Publications

A Unifying Framework for the v -Tamari Lattice and Principal Order Ideals in Young's Lattice

Published in **Combinatorica Journal (Springer)** – June 2023

On Framed Triangulations of Flow Polytopes, the v -Tamari Lattice, and Young's Lattice

Final Version: **Apr 2021** – Bar-Ilan University, Ramat Gan, Israel

Published in **Séminaire Lotharingien de Combinatoire, FPSAC 2021 Proceedings**

Activities & Events

- **Regional Programming Marathon (South America North) ICPC** – Participant, Politécnico Grancolombiano (Nov 2019)
- **XXII Colombian Mathematics Congress** – Speaker, Universidad del Cauca, Popayán (Jun 2019)
 - **Poster:** Flow Polytopes and Triangulations of the Caracol Polytope
- **Cube Challenge Sergista** – Organizer, Universidad Sergio Arboleda (2018 & 2019)